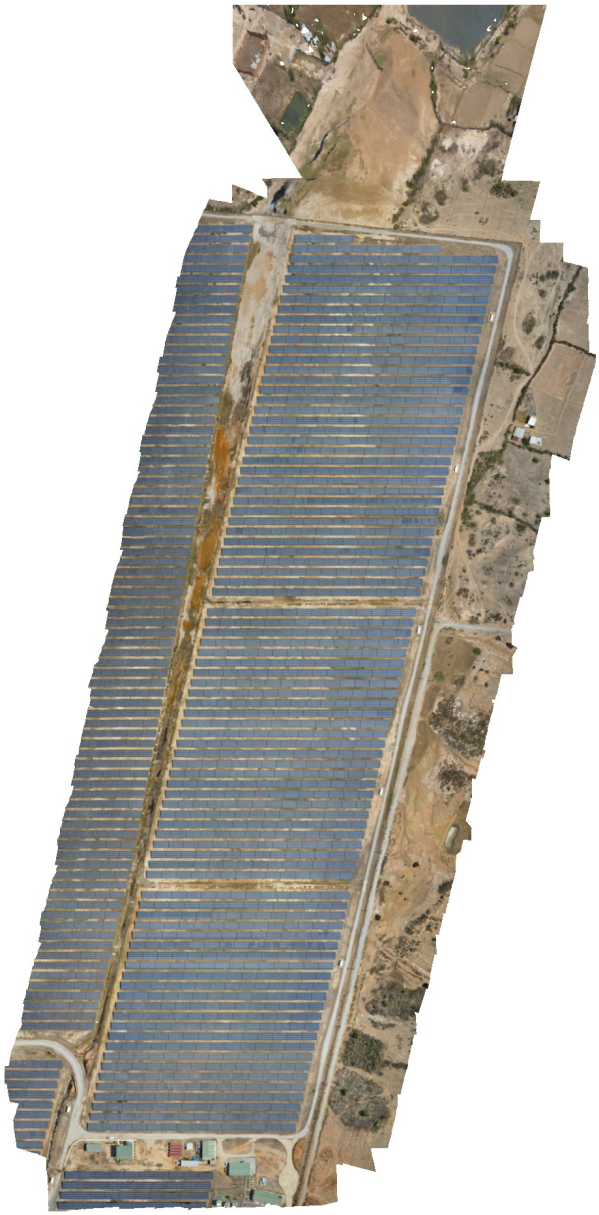
Agisoft Metashape

Processing Report

08 September 2026



Survey Data

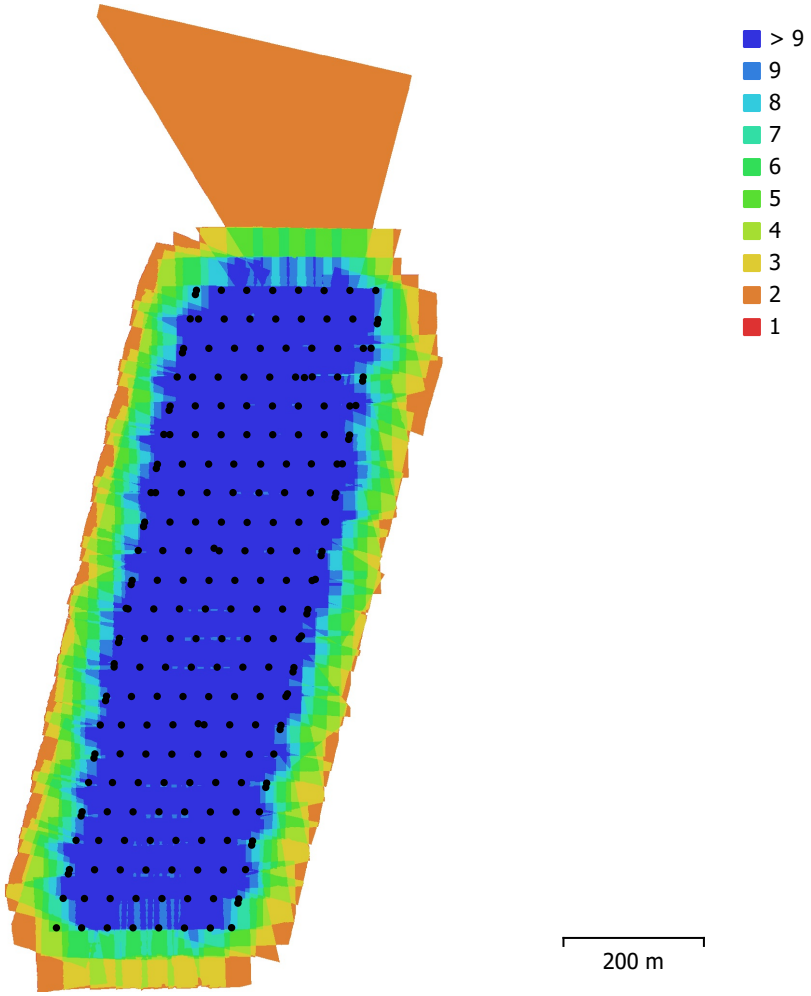


Fig. 1. Camera locations and image overlap.

Number of images:	222	Camera stations:	222
Flying altitude:	125 m	Tie points:	127,642
Ground resolution:	2.02 cm/pix	Projections:	440,918
Coverage area:	0.538 km²	Reprojection error:	0.753 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
MAVIC2-ENTERPRISE-A...	8000 x 6000	4.5 mm	0.811 x 0.811 µm	No

Table 1. Cameras.

Camera Calibration

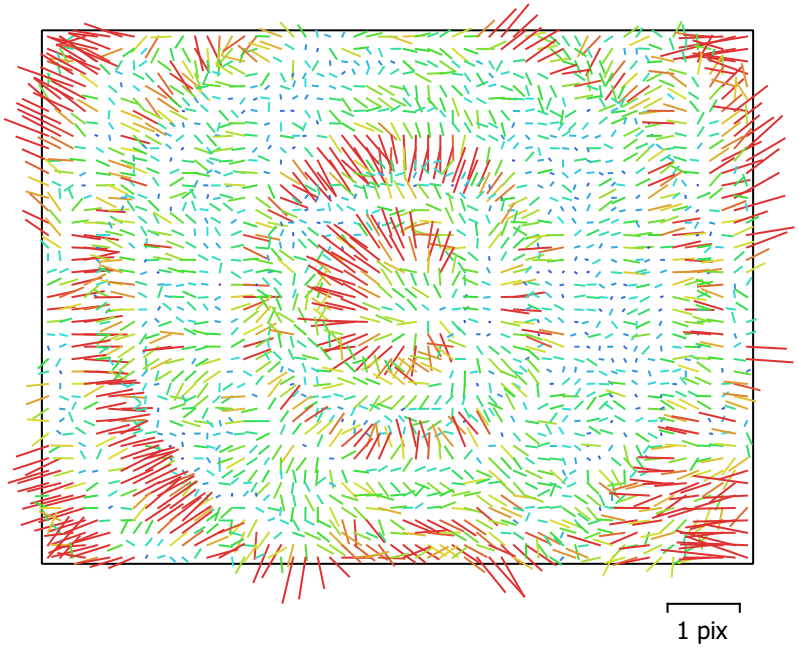


Fig. 2. Image residuals for MAVIC2-ENTERPRISE-ADVANCED (4.5mm).

MAVIC2-ENTERPRISE-ADVANCED (4.5mm)

222 images

Type	Resolution	Focal Length	Pixel Size
Frame	**8000 x 6000**	**4.5 mm**	**0.811 x 0.811 µm**

	Value	Error	F	Cx	Cy	B1	B2	K1	K2	K3	P1	P2
F	**5885.97**	1.1	1.00	0.46	0.24	0.04	-0.17	0.32	-0.16	0.07	-0.04	0.13
Cx	**12.4242**	0.27		1.00	0.18	0.00	-0.35	0.02	-0.07	0.04	-0.01	-0.01
Cy	**17.7687**	0.2			1.00	0.26	-0.09	0.03	-0.04	0.02	0.01	0.06
B1	**-2.18782**	0.015				1.00	0.05	-0.04	-0.07	0.07	-0.05	-0.07
B2	**-1.26919**	0.015					1.00	-0.01	0.02	-0.01	0.07	0.01
K1	**0.000344855**	6.9e-05						1.00	-0.59	0.50	0.12	0.38
K2	**-0.0246095**	0.00012							1.00	-0.98	0.03	-0.03
K3	**0.00772831**	0.00012								1.00	-0.03	-0.00
P1	**1.71095e-06**	2.6e-06									1.00	0.12
P2	**6.64256e-05**	2.3e-06										1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

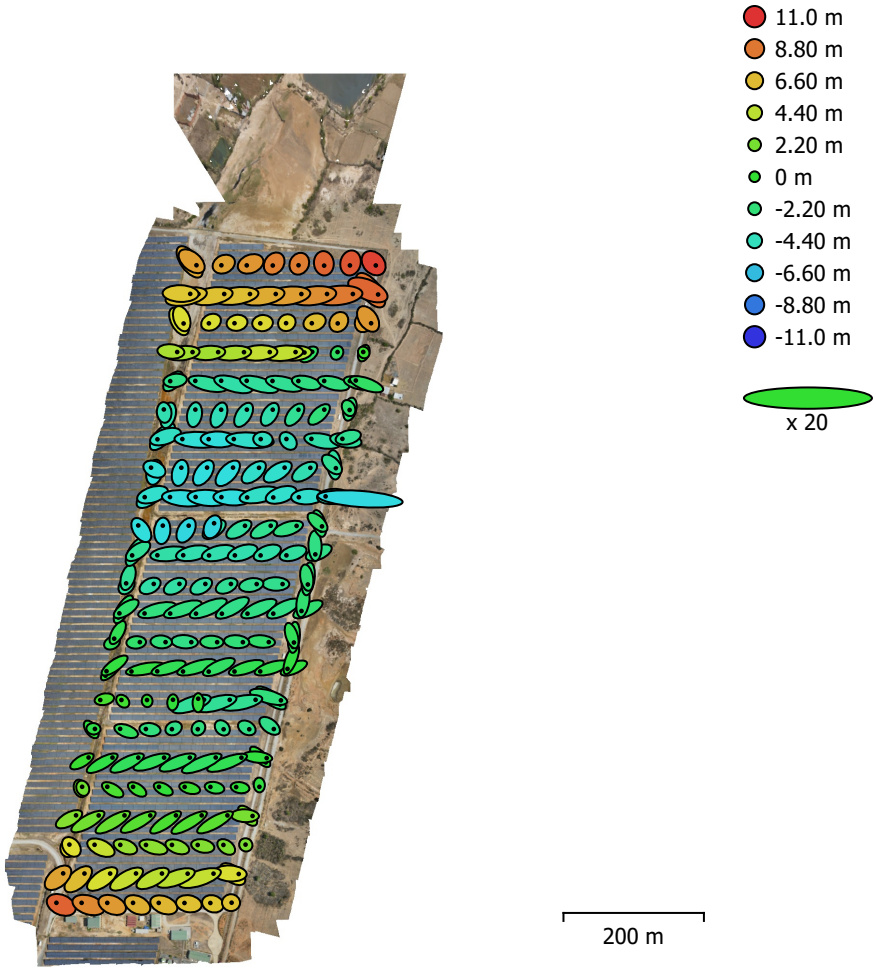


Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (m)	Y error (m)	Z error (m)	XY error (m)	Total error (m)
1.13509	0.552812	4.30041	1.26255	4.48192

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

Digital Elevation Model

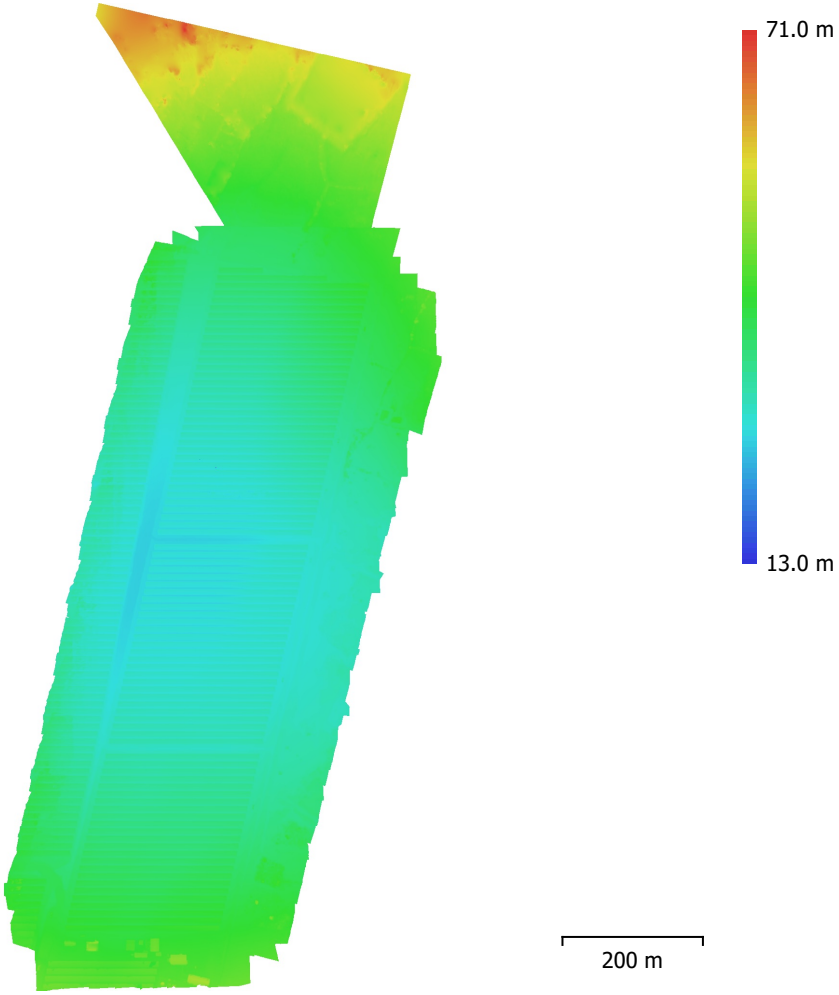


Fig. 4. Reconstructed digital elevation model.

Resolution:	4.23 cm/pix
Point density:	559 points/m²

Processing Parameters

General	
Images	222
Aligned images	222
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll
Tie Points	
Points	127,642 of 272,412
RMS reprojection error	0.235839 (0.75259 pix)
Max reprojection error	2.5117 (8.95526 pix)
Mean key point size	3.36225 pix
Point colors	3 bands, uint8
Key points	730.00 MB
Average tie point multiplicity	3.29917
Alignment parameters	
Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	Yes
Matching time	6 minutes 1 seconds
Matching memory usage	381.57 MB
Alignment time	2 minutes 14 seconds
Alignment memory usage	239.25 MB
Optimization parameters	
Parameters	f, cx, cy, k1, k2, p1, p2
Adaptive camera model fitting	Yes
Exclude corners	No
Optimization time	3 seconds
Date created	2026:09:05 01:21:59
Software version	2.3.1.22580
File size	750.88 MB
Depth Maps	
Count	222
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	44 minutes 47 seconds
Memory usage	8.15 GB
Date created	2026:09:05 02:15:45
Software version	2.3.1.22580
File size	3.58 GB
Point Cloud	
Points	314,981,504
Coordinate precision	1.01 cm
Point attributes	

Color	3 bands, uint8
Normal	
Point classes	
Created (never classified)	314,981,504
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	44 minutes 47 seconds
Memory usage	8.15 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	51 minutes 28 seconds
Memory usage	10.67 GB
Date created	2026:09:05 03:07:22
Software version	2.3.1.22580
File size	4.07 GB
DEM	
Size	14,698 x 33,216
Resolution	4.23 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	44 minutes 47 seconds
Memory usage	8.15 GB
Reconstruction parameters	
Source data	Depth maps
Interpolation	Enabled
Processing time	1 hours 44 minutes
Memory usage	8.18 GB
Date created	2026:09:05 04:58:25
Software version	2.3.1.22580
File size	1.24 GB
Orthomosaic	
Size	29,397 x 66,432
Resolution	2.12 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Colors	3 bands, uint8
Orthophotos	2.91 GB
Reconstruction parameters	
Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	10 minutes 26 seconds
Memory usage	4.09 GB
Date created	2026:09:05 07:21:34
Software version	2.3.1.22580
File size	3.82 GB
System	
Software name	Agisoft Metashape Professional
Software version	2.3.0 build 21954
OS	Windows 64 bit
RAM	15.78 GB

CPU	11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz
GPU(s)	NVIDIA GeForce RTX 3050 Laptop GPU